JAMES MCCALLUM

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PROFESSIONAL SUMMARY

I'm a Staff Engineer with 10 years of experience building scalable backend systems and leading technical initiatives across fintech and SaaS sectors. I've spent the last four years in staff-level roles focusing on infrastructure modernisation, mentoring engineering teams, and solving complex architectural challenges. My strength lies in taking ambiguous problems and breaking them down into clear technical strategies that drive measurable business impact.

WORK EXPERIENCE

Staff Engineer

Cascade Technology Ltd

cascade-tech.io

Edinburgh, United Kingdom

March 2022 - Present

Designed and implemented a microservices migration strategy that reduced API response times by 40% and improved system reliability from 99.5% to 99.95% uptime across 12 services handling 2 million daily requests.

Led a team of 5 senior engineers through the adoption of event-driven architecture using Kafka, resulting in decoupled systems that accelerated feature delivery velocity by 35% and reduced deployment incidents by 60%.

Established engineering standards for observability including distributed tracing with Jaeger and custom metrics collection, enabling faster incident resolution and reducing mean time to recovery from 45 minutes to 12 minutes.

Mentored 8 engineers across multiple teams, with 3 being promoted to senior positions within 18 months, and created a knowledge sharing program that decreased onboarding time by 6 weeks for new joiners.

Senior Backend Engineer

DataFlow Systems

dataflowsystems.com

Edinburgh, United Kingdom

June 2018 - February 2022

Built and maintained a real-time data processing platform handling 500 million events daily using Go and PostgreSQL, serving 40+ enterprise clients with 99.9% uptime guarantees.

Optimised database query performance across 20 slow-running reports by implementing query refactoring and strategic indexing, reducing average query time from 8 seconds to 400 milliseconds.

Drove the adoption of containerisation and Kubernetes for all backend services, automating deployment processes and reducing infrastructure provisioning time from 2 days to 15 minutes.

Collaborated with the product team to design and deliver a customer-facing REST API used by 500+ third-party integrations, which became the foundation for 25% revenue growth in the platform licensing segment.

Software Engineer

NetLogic Solutions

netlogicsolutions.net

Edinburgh, United Kingdom

August 2014 - May 2018

Developed core features for a cloud-based network monitoring platform using Java and Spring Boot, contributing to systems used by 100+ enterprise customers across EMEA region.

Implemented comprehensive automated testing framework covering unit, integration, and end-to-end tests, increasing code coverage from 45% to 78% and reducing production bugs by 50%.

Participated in on-call rotation for production support, resolving critical incidents affecting customer infrastructure and maintaining 99.7% service availability.

Maintained and refactored legacy C++ components for customer data ingestion pipelines, improving throughput by 25% and reducing technical debt blocking new feature development.

EDUCATION

Bachelor of Science in Computer Science

University of Edinburgh

Edinburgh, United Kingdom

2010 - 2014

TECHNICAL SKILLS

Languages: Go, Java, Python, SQL, Bash, JavaScript

Databases: PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch

Cloud and Infrastructure: AWS (EC2, S3, RDS, Lambda), Kubernetes, Docker, Terraform, Prometheus

Architectures: Microservices, Event-driven systems, RESTful APIs, Message queues (Kafka, RabbitMQ)

Tools and Platforms: Git, CI/CD pipelines (Jenkins, GitHub Actions), Jaeger, Grafana, DataDog

Methodologies: Agile, Code review, System design, Performance optimization

GITHUB REPOSITORIES

github.com/jamesmccallum/query-optimizer - Database query optimization library with benchmarking tools

github.com/jamesmccallum/event-processor - Go library for building event-driven applications with Kafka

github.com/jamesmccallum/observability-toolkit - Distributed tracing and metrics collection utilities

github.com/jamesmccallum/container-utils - Docker image building and optimization utilities